```
-------------------------------------------------------------------------------------
> -----------------------------------------------------------------
      name:  <unnamed>
       log:  F:\ECON 408\Practice do file\Homework-2.log
  log type:  text
 opened on:  14 May 2026, 19:45:30

.
. * (1) Biased ARMA(2,1) Model Simulation
. *--------------------------------------------------------------------------------
.
. clear all

. set seed 1000

.
. capture postclose buffer

. postfile buffer beta_L1 beta_L2 alpha_L1 using arma22_misspecified.dta, replace

.
. forvalues i = 1/1000 {
  2.     qui drop _all
  3.         qui set obs 500
  4.         qui gen time = _n
  5.         qui tsset time
  6.
.         qui gen double e = rnormal()
  7.         qui gen double y = e in 1
  8.         qui replace y = 0.4*L1.y + e + 0.5*L1.e in 2
  9.         qui replace y = 0.4*L1.y + 0.3*L2.y + e + 0.5*L1.e + 0.3*L2.e in 3/L
 10.         qui arima y, ar(1/2) ma(1)
 11.         post buffer (_b[ARMA:L1.ar]) (_b[ARMA:L2.ar]) (_b[ARMA:L1.ma])
 12.
. }

.
. postclose buffer

. use arma22_misspecified.dta,clear

.
. sum beta_L1 beta_L2 alpha_L1

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
     beta_L1 |      1,000    1.324907    .5218545  -.0861322   1.883718
     beta_L2 |      1,000   -.4230766    .4589573  -.8911305   .8230543
    alpha_L1 |      1,000   -.4310591    .4753887   -1.13755     .86856

.
. set scheme white_tableau

. graph set window fontface "Arial Narrow"

.
. * AR(1) Plot
. kdensity beta_L1, lcolor(navy) lpattern(solid) lwidth(medthick) ///
>          xline(0.4, lcolor(black) lpattern(shortdash)) ///
>                  name(g1, replace) title("{bf:AR(1) Parameter}") ///
>                  xtitle("Estimated Value") ///
>                  xlabel(0 0.4 0.5 1 1.5 2 ,labsize(medium) nogrid) ///
>                  ylabel(, labsize(medium) nogrid) note(" ")
```

```
.
. * AR(2) Plot
. kdensity beta_L2, lcolor(maroon) lpattern(solid) lwidth(medthick) ///
>          xline(0.3, lcolor(black) lpattern(shortdash)) ///
>                  name(g2, replace) title("{bf:AR(2) Parameter}") ///
>                  xtitle("Estimated Value") ///
>                  xlabel(-1 -0.5 0 0.3 0.5 1 ,labsize(medium) nogrid) ///
>                  ylabel(, labsize(medium) nogrid) note(" ")

.
. * MA(1) Plot
. kdensity alpha_L1, lcolor(forest_green) lpattern(solid) lwidth(medthick) ///
>          xline(0.5, lcolor(black) lpattern(shortdash)) ///
>                  name(g3, replace) title("{bf:MA(1) Parameter}") ///
>                  xtitle("Estimated Value") xlabel( ,labsize(medium) nogrid) ///
>                  note(" ") ylabel(, labsize(medium) nogrid)

.
. * Combine Graph
. graph combine g1 g2 g3, ///
>       rows(1) title("{bf:Simulation Results: ARMA(2,1)}") ///
>           xsize(8) ysize(4) ///
>           note("Black dashed line represents the true parameter") ///
>           name(Biased, replace)

.
. graph export Biased_arma.emf, name(Biased) replace
file F:\ECON 408\Practice do file\Biased_arma.emf saved as Enhanced Metafile format

.
.
.
. * (2) Unbiased Correct ARMA(2,2) Model Simulation
. *------------------------------------------------------------------------------
.
. clear all

. set seed 1000

.
. capture postclose buffer

. postfile buffer beta_L1 beta_L2 alpha_L1 alpha_L2 using arma22_correct_model.dta, re
> place

.
. forvalues i = 1/1000 {
  2.     qui drop _all
  3.         qui set obs 500
  4.         qui gen time = _n
  5.         qui tsset time
  6.
.         qui gen double e = rnormal()
  7.         qui gen double y = e in 1
  8.         qui replace y = 0.4*L1.y + e + 0.5*L1.e in 2
  9.         qui replace y = 0.4*L1.y + 0.3*L2.y + e + 0.5*L1.e + 0.3*L2.e in 3/L
 10.         qui arima y, ar(1/2) ma(1/2)
 11.         post buffer (_b[ARMA:L.ar]) (_b[ARMA:L2.ar]) (_b[ARMA:L.ma]) (_b[ARMA:L2.
> ma])
 12.
```

```
. }

. 
. postclose buffer

. use arma22_correct_model.dta,clear

. 
. sum beta_L1 beta_L2 alpha_L1 alpha_L2

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
     beta_L1 |      1,000    .4144489    .1664455  -.0255998   1.370499
     beta_L2 |      1,000    .2762445    .1460104  -.4907512   .6360206
    alpha_L1 |      1,000    .4834226    .1622318  -.4647181   .8953954
    alpha_L2 |      1,000    .3050435    .0505966   .1358287   .4744434

. 
. set scheme white_tableau

. graph set window fontface "Arial Narrow"

. 
. 
. * AR(1) Plot
. kdensity beta_L1, lcolor(navy) lpattern(solid) lwidth(medthick) ///
>     xline(0.4, lcolor(black) lpattern(shortdash)) ///
>     name(g1, replace) title("{bf:AR(1) Parameter}") ///
>     xtitle("Estimated Value" ,size(medium)) ytitle(, size(medium)) note(" ") ///
>     xlabel(0 0.4 0.5 1 1.5, labsize(medium) nogrid) ylabel(, labsize(medium) nogrid)

. 
. * AR(2) Plot
. kdensity beta_L2, lcolor(maroon) lpattern(solid) lwidth(medthick) ///
>     xline(0.3, lcolor(black) lpattern(shortdash)) ///
>     name(g2, replace) title("{bf:AR(2) Parameter}") ///
>     xtitle("Estimated Value", size(medium)) ytitle(,size(medium)) note(" ") ///
>     xlabel(-0.5 0 0.3 0.5, labsize(medium) nogrid) ylabel(, labsize(medium) nogrid)

. 
. * MA(1) Plot
. kdensity alpha_L1, lcolor(forest_green) lpattern(solid) lwidth(medthick) ///
>     xline(0.5, lcolor(black) lpattern(shortdash)) ///
>     name(g3, replace) title("{bf:MA(1) Parameter}") ///
>     xtitle("Estimated Value", size(medium)) ytitle(,size(medium)) note(" ") ///
>     xlabel(-0.5 0 0.5 1, labsize(medium) nogrid) ylabel(, labsize(medium) nogrid)

. 
. * MA(2) Plot
. kdensity alpha_L2, lcolor(purple) lpattern(solid) lwidth(medthick) ///
>     xline(0.3, lcolor(black) lpattern(shortdash)) ///
>     name(g4, replace) title("{bf:MA(2) Parameter}") ///
>     xtitle("Estimated Value",size(medium)) ytitle(,size(medium)) note(" ") ///
>     xlabel(0 0.3 0.5, labsize(medium) nogrid) ylabel(, labsize(medium) nogrid)

. 
. * Combine into 2x2 Grid
. graph combine g1 g2 g3 g4, ///
>     rows(2) cols(2) ///
>     title("{bf:Simulation Results: Correctly Specified ARMA(2,2)}") ///
>     note("Black dashed line represents the true parameter", size(small)) ///
>     xsize(8) ysize(4) name(CorrectModel, replace)
```

```
.
.
. graph export ARMA_Correct.emf, name(CorrectModel) replace
file F:\ECON 408\Practice do file\ARMA_Correct.emf saved as Enhanced Metafile format

.
.
.
. log close
      name:  <unnamed>
       log:  F:\ECON 408\Practice do file\Homework-2.log
  log type:  text
 closed on:  14 May 2026, 19:55:07
--------------------------------------------------------------------------------
> ----------------------------------------------------------------
```